

Comparative efficacy of *Annona squamosa* and *Azadirachta indica* extracts against *Boophilus microplus* Izatnagar isolate

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Abstract In the search of developing herbal acaricides, eight medicinal plants were screened for their efficacy against *Boophilus microplus*, the widely distributed tick species in India. Of the seven extracts screened, the extracts prepared from the *Annona squamosa* seed showed very high level of efficacy (70.8%) after 24 h of treatment. The effect of treatment on oviposition of the survived ticks was also assessed, and a significant reduction ($P < 0.05$) in the reproductive index was noted in comparison to control. When efficacy of the in vitro optimized concentration of *A. squamosa* was compared with previously tested extract of *Azadirachta indica* in in vivo model, it was observed that the extracts prepared from *A. indica* is more efficacious than the extracts of *A. squamosa*. A comparable efficacy against *B. microplus* fed on animals treated with herbal extracts and commonly used synthetic acaricide was noted. The possibility of using the herbal extracts in IPM format for the management of ticks is discussed.

Introduction

In India, *Boophilus microplus* tick is widely distributed and infesting several host species and economically impact cattle production by reducing weight gain, milk production,

and also by transmitting pathogens that cause babesiosis and anaplasmosis (Ghosh et al. 2006). As per the latest estimate, the control cost of ticks and tick-borne diseases (TTBDs) in India has been determined in the tune of about US\$498.7 million per annum (Minjauw and McLeod 2003). Keeping in view of the impact of TTBDs on animal productivity, control of tick infestations has been given priority for the improvement of animal health. Presently, tick control is focused on the repeated use of organophosphates and pyrethroid group of acaricides. However, application of chemical acaricides has had limited efficacy in reducing tick infestations and is often accompanied by serious drawbacks, including the development of acaricide resistant ticks, environmental contamination, and even contamination of milk and meat products with insecticide residues (Graf et al. 2004). As per the FAO (2004) report, Indian ticks have developed resistance to commonly used acaricides. During recent surveys, farmers of some of the Indian states were interacted, and acaricide resistance problem has been noted (Ghosh, personal observation). To tackle the problem, concerted attempts have been directed to develop suitable herbal formulation for tick control through National Agricultural Innovation Project (<http://www.tickcontrolivri.com>). In the present experiment, an attempt was made for in vitro screening of some of the medicinal plant extracts against *B. microplus* and for evaluating the efficacy of the effective extract in vivo.

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Materials and methods

Maintenance of *Babesia bigemina* free *B. microplus* colony

For the development of integrated control program against economically important tick species of Indian livestock, *B.*

microplus and *Hyalomma anatolicum anatolicum* are continuously maintained in the Entomology Laboratory of the Division of Parasitology, IVRI as per the method of Khan et al. (1982) and Ghosh and Azhahianambi (2007). The laboratory-reared ticks were used for in vitro and in vivo experiments.

Animals

Sixteen healthy cross-bred calves of similar age group (6–7 months of age) were procured from the Livestock Product Management section of the institute and were reared in tick proof shed of the Division of Parasitology under standard feeding and management condition. All the animals were kept on dry fodder to avoid extraneous infestation of ticks through green fodder. The tick-free status of the calves were maintained by periodical application of 0.05% cypermethrin on the host and 0.1% chlorpyrifos on animal shed. The insecticide application was suspended 45 days before initiation of the experiment. Animals were randomly divided into three groups (I, II, and III) of five animals in each group.

Animal experimentation was conducted in accordance with the approved guidelines laid down by the Committee for the Purpose of Control and Supervision of Experimentation on Animals (CPCSEA), a monitoring Indian statutory body.

Plant/herb materials

In the present study, seed of “*Annona squamosa*, *Azadirachta indica*, and *Tamarindus indicus*, leaf of *Nicotiana tobacum*, *Eucalyptus globules*, and *Citrus leminum*, bulb of *Allium sativum*, and *Withania somnifera* were selected on the basis of available literature and on ethnoveterinary knowledge of rural people. The taxonomic position of the plants were identified from the Department of Botany, Bareilly College, Rohailkhand University, Bareilly, India. Different parts of the collected plant/herb materials were shed-dried for 15 days.

Extraction procedure

The selected plant materials were shade-dried and crushed in grinder to make a powdery consistency. One hundred gram each of the crushed material was positioned separately in a columnar extraction system of Soxhlet apparatus, and 250 ml of ethanol at 80°C was provided for extraction. The extraction process was carried out up to 12–15 cycles. Recovered extracts were passed through a rotatory evaporator to remove the solvent. The extract was dried under vacuum below 40°C. The semi-dried contents were collected and weighed. The yields of different plant materials were recorded.

In vitro study

Effect on mortality

Extracts were screened for acaricidal property against *B. microplus* by employing tick immersion method recommended by Benavides et al. (2001). For initial screening of plant extracts, the engorged female ticks were washed thoroughly thrice with distilled water and kept for drying on filter paper. After drying ticks ($n=6 \times 5$ replications, including control) were immersed separately in an 8% concentration of each extract, whereas control ticks were immersed in distilled water for 2 min and then transferred to the test Petri dishes padded with Whatmann filter paper no. 1. All the Petri dishes were kept at 28°C and $85 \pm 2\%$ relative humidity in desiccators. The effect of the extracts on tick mortality was recorded at 1, 2, 3, 5, and at 24 h of treatment. Dead ticks were confirmed by observing loss of motility, pedal reflex after exposing to light, and also by loss of oviposition property. LC_{50} of the most effective plant extract found in the preliminary screening test was determined by the concentration dependent mortality of adult female *B. microplus* ticks. For each concentration, 30 preweighed ticks ($n=6 \times 5$ replication) were taken. Extracts of various concentrations of 1–8% were prepared, individual tick group was subjected to tick immersion test, and mortality data were recorded as mentioned above.

Effect on reproductive index

The ticks survived after exposing to 1–8% concentration of extract were reared subsequently for generating the data on effect of most effective plant extract on reproductive index (RI = egg mass/engorged tick weight). Survived ticks were individually weighed and kept inside the glass specimen tubes and covered with muslin cloth. All the tubes were kept in desiccators under optimum environment (28°C and $85 \pm 2\%$ RH) for oviposition. When oviposition ceased, the eggs laid in each tube was weighed separately. The optimum dose of most effective plant extract was determined on the basis of highest percent mortality and lower RI and was tested in in vivo trial.

Safety analysis

The selected plant extract was subjected to toxicity trial as per the guidelines of WHO (WHO 1993). In vitro optimized concentration of plant extract was applied topically to finely shaved and demarcated skin area of rabbit ($n=4$). The skin thickness was recorded before and 24 h after application. Any changes at the site of application of the extract like staining, pruritis, and erythematic lesion

Table 1 Details of plant extracts screened, their common name, solvent used in extraction process, and yield percentage

Plants	Common name	Solvent used	Yield (g%)
Seed <i>Annona squamosa</i>	Sitaphal	Ethanol	18.7
seed <i>Tamarindus indicus</i>	Imli	Ethanol	19.3
Leaf <i>Nicotiana tobacum</i>	Tambacu	Ethanol	16.1
leaf <i>Eucalyptus globulus</i>	Nilgiri	Ethanol	28.4
Leaf <i>Citrus leminum</i>	Lemon	Ethanol	17.5
Garlic bulb <i>Allium sativum</i>	lasoon	Ethanol	8.1
Ashwagandha <i>Withania somnifera</i>	Ashwagandha	Ethanol	14.8

were recorded at different intervals. To rule out the toxic effect of the extracts completely, rabbits ($n=4$) were treated with 16% and 32% concentration of the extract, and data were recorded.

In vivo study

All the calves of groups I–IV were experimentally challenged with 5–7-day-old larvae of *B. microplus* (each animal was challenged by larvae generated from 50 mg freshly laid eggs) by ear bag method (Ghosh and Azhahianambi 2007). The ear bags were checked daily. Eighteen days of experimental challenge, calves of group I were treated with water, and they served as infected control group (IC). Calves of group II were treated with 0.05% concentration of cypermethrin and were served as positive control group (PC). Animals of group III (T_1) were treated topically with in vitro optimized concentration of extracts of *A. squamosa*. Animals of group IV (T_2) were treated with previously tested 8% concentration (Srivastava et al. 2008) of neem seed extract. The application was done by cotton swab. At the end of the experiment, all the animals were treated with chemical acaricide to keep them tick-free (as per the approved guidelines of CPCSEA). After treatment, ticks were collected from the ear bag of all the animals and were categorized. The live ticks from all the animals were weighed individually and kept in tick rearing

tube for oviposition. The RI, alteration in the egg mass production (DO%), alteration in the weight of adult tick (DR%), and the percentage efficacy of the extract (E%) (Ghosh et al. 2005) of ticks from all the groups of animals were calculated as follows:

Reproductive index (RI) = egg mass/engorged tick weight

DR % (% reduction of mean weight of adult female) = $100(1 - \text{PMTV}/\text{PMTC})$ where PMTV = mean weight of adult female of treated group and PMTC = mean weight of adult female of control group.

DO % (% reduction of mean weight of eggs) = $100(1 - \text{PATV}/\text{PATC})$ where, PATV = mean weight of eggs laid by ticks of treated group and PATC = mean weight of eggs laid by ticks of control group.

E % (Efficacy of drug) = $E\% = 100[1 - (\text{CRT} \times \text{CRO})]$ where CRT = reduction in weight of adult female PMTV/PMTC and CRO = reduction in egg laying capacity PATV/PATC.

Statistical analysis

The LC_{50} value was determined by probit analysis (Finney 1962), while differences in mean values of entomological data among the groups were analyzed by Student's *t* test (Snedecor and Cochran 1968).

Table 2 Percent mortality of *B. microplus* ticks following in vitro screening of various plant extract

Plant materials	1st hour	2nd hour	3rd hour	5th hour	24th hour
<i>Annona squamosa</i>	12.5	29.2	33.3	54.2	70.8
<i>Nicotiana tobacum</i>	08.3	16.7	29.2	37.5	45.8
<i>Tamarindus indica</i>	08.3	16.7	25.0	37.5	41.7
<i>Eucalyptus globulus</i>	0	12.5	12.5	16.7	29.8
<i>Citrus lemon</i>	0	0	08.3	12.5	20.8
<i>Withania somnifera</i>	–	04.2	12.5	16.78	16.7
<i>Allium sativum</i>	0	0	4.2	4.2	08.3

Table 3 In vitro evaluation acaricidal property of *A. squamosa* at various concentrations against *B. microplus*

Concentration	1st hour	2nd hour	3rd hour	5th hour	24th hour
8%	12.5	29.1	33.3	54.6	70.8
7%	12.5	16.6	33.3	45.8	58.3
6%	08.3	16.6	25.0	37.5	41.6
5%	04.1	12.5	20.8	33.3	41.6
4%	04.1	04.1	16.6	25.0	37.5
3%	0	04.1	12.5	12.5	20.8
2%	0	0	08.3	08.3	12.5
1%	0	0	0	0	04.1
Control	0	0	0	0	0

Results

Screening of plants and yield percentage

The scientific names, common names of the plants screened, solvent used for the extraction process, and yield percentages are given in Table 1.

In vitro study

It was observed that, after 24 h of treatment, the highest acaricidal activity of 70.8% was recorded in the ticks treated with 8% extract of *A. squamosa* followed by *N. tobacum* (45.8%) and *T. indica* (41.7%) extracts, while 29.8% and 20.8% mortality, respectively, was recorded in ticks treated with *E. globulus* and *C. leminum* extracts. A comparatively lower efficacy of 16.7% and 8.3% was recorded in ticks treated with *W. somnifera* and *A. sativum* extracts, respectively (Table 2).

On the basis of tick mortality pattern within 24 h of treatment, the LC₅₀ value was determined as 5.62. From dose-dependent response data, it is very clear that mortality is directly linked with the dose applied, and a minimum of 8% concentration is required to achieve more than 70% mortality (Table 3).

By comparing RI of ticks exposed with different concentration of *A. squamosa* extract, it was observed that the ticks

treated with 8% concentration laid fewer eggs, and the data were significantly ($P<0.05$) different from the data generated from the ticks treated with other concentrations except 7% (Table 4). Critical analysis of the results revealed that, although there was no significant difference in the RI of the ticks treated with 8% (0.26 ± 0.003) and 7% (0.25 ± 0.005) concentrations, there was a remarkable difference in terms of mortality percentage of ticks treated with 7% and 8% concentrations (Table 3). There was a significant difference ($P<0.05$) between the RI of the treated ticks and the ticks from the control group. From the data generated in the in vitro study, 8% concentration was selected for in vivo study

Skin reaction

There was no change in the skin thickness of rabbit treated with 8% concentration after 24 h of topical application of *A. squamosa* extract. It was encouraging to note that rabbits treated with 16% and 32% concentrations of extract did not show any adverse reactions. Other parameters like staining, pruritis, and erythematic lesion were also not observed at the site of application.

In vivo study

After topical application of extract, immediate effect on ticks and its knockdown period were recorded in all the

Table 4 Reproductive indices of *B. microplus* treated with various concentrations of *A. squamosa* extract

Concentration (%)	Engorgement weight of ticks (mg)	Egg weight (mg)	Reproductive index (RI)
8	145.0±0.58	37.84±0.42	0.26±0.003 ^a
7	139.0±0.67	35.02±0.58	0.25±0.005 ^a
6	128.0±1.70	41.97±1.33	0.33±0.001 ^b
5	114.0±1.16	33.62±1.54	0.29±0.008 ^c
4	143.0±0.54	45.33±1.73	0.32±0.001 ^{bde}
3	120.0±0.58	38.48±1.90	0.32±0.001 ^{bdef}
2	147.0±0.53	48.43±4.01	0.33±0.001 ^{bdef}
1	109.0±1.45	43.14±0.80	0.39±0.009 ^g
Control	140.0±1.152	54.32±1.03	0.39±0.001 ^h

Observation with superscript letters in a column differ significantly at the level of $P<0.05$

Table 5 Reproductive indices of *B. microplus* ticks subjected to different treatments during in vivo study

Groups	Tick weight (mg)	Egg weight (mg)	Reproductive index (RI)
I	135.2±2.7 ^a	43.1±2.1 ^a	0.32±0.001 ^a
II	105.6±6.1 ^b	20.2±2.8 ^{ab}	0.19±0.003 ^b
III	116.93±3.71 ^{bc}	34.5±5.2 ^{bc}	0.30±0.005 ^{ab}
IV	114.4±3.9 ^{bcd}	27.6±5.4 ^c	0.23±0.004 ^b

Observation with superscript letters in a column differ significantly at the level of $P<0.05$

treated groups (groups II, III, and IV). Among the ticks fed on animals of group II, 86.4% dropped within 3 days post-treatment (dPT) and died without laying eggs, while 65.9% and 71.8%, respectively, of the ticks fed on animals of groups III and IV dropped within 6 and 5 dPT and were died without laying eggs (Table 6). Ticks fed on group I calves dropped out 10th dPT.

The mean weight of the ticks fed on animals of groups II, III, and IV were 105.6, 116.9, and 114.4 mg, respectively, while the same for control tick was 135.2 mg, and the data was differed significantly ($P<0.05$). The mean egg mass produced from the ticks of groups II (CT), III (HT₁), and IV (HT₂) were 20.2, 34.5, and 27.6 mg, respectively, whereas control ticks laid 43.1 mg. The reduction in the egg masses of 22.9, 8.6, and 15.5 mg of ticks fed on treated groups II, III, and IV, respectively, was significantly ($P<0.05$) differ from the egg mass laid by the ticks fed on animals of group I. Mean reproductive indices of ticks fed on groups II (CT), III (HT₁), and IV (HT₂) animals were 0.19, 0.30, and 0.23, respectively, and the value was significantly ($P<0.05$) differed from the ticks collected from the animals of group I (Table 5). The DR%, DO%, and E% of ticks fed on the animals of groups II, III, and IV were calculated and tabulated (Table 6).

Discussion

In our continuous efforts to develop sustainable integrated control measures against multitick infestations in dairy and food animals, attempts were made to identify plants/herbs having acaricidal properties. Seven different plants having medicinal properties were selected, and extracts were prepared. Of the seven extracts prepared, highest yield of 28.4% was recorded from the leaf of *E. globules*, while 8.1% yield was recorded from bulb of *A. sativum*. Following in vitro experimentation, it was observed that, at 8% concentration of extracts, 70.8% of the ticks treated

with *A. squamosa* died within 24 h of treatment, while *A. sativum* gave only 8.3% efficacy against *B. microplus* (Table 2). Previously, Narang et al. (1988) reported that ethanolic extract from *A. squamosa* seed has more larvicidal activity than from leaves and also recorded the high acaricidal effect of polar solvent extract of *A. squamosa* on *B. microplus* within 24 h after dipping.

Sustainable tick control strategy emphasizes that the control should include both on and off the host (Nari 1995). Thus, population-limiting property of any plant extract need to be evaluated to assess the efficacy of extracts. In the present study, the effects of *A. squamosa* seed extract on the future progeny of the exposed ticks were evaluated by estimating the reproductive index (Ghosh et al. 2005), and the data revealed a significant ($P<0.05$) reduction in the egg-laying capacity of all the ticks exposed to different concentrations (3–8% concentration) of *A. squamosa* in comparison to the ticks of control group. Previously, Srivastava et al. (2008) assessed the population-limiting potential of neem seed extract, and a significant reduction of 0.34% in the RI of the ticks treated with 8% concentration of the extract in comparison to the ticks of control group was recorded. Similarly, Kalakumar et al. (2000) reported significant effect of commercially available neem oil and custard oil on egg-laying capacity of *H. anatolicum anatolicum*, *Rhipicephalus haemophysaloides*, and *B. microplus*.

Before conducting in vivo trial, the safety level of the extracts was evaluated and found safe. However, Nongluck et al. (2006) reported different level of toxicity of the *A. squamosa* extracts prepared in petroleum ether, ether, chloroform, and ethanol. Results revealed that some extracts caused conjunctival redness, chemosis, rugged cornea, skin erythema, and edema. The organic solvent extracts that produced adverse effects on skin of rabbit eyes were in the following order of severity: ether > petroleum ether > chloroform > ethanol, and those on rabbit ear skin

Table 6 Mortality %, knockdown period, DR%, DO%, and E% of *B. microplus* ticks subjected to different treatments during in vivo study

Group	Mortality %	Knockdown period	DR%	DO%	E%
II	86.4	3	22.2	69.4	76.1
III	65.9	6	13.3	19.8	31.2
IV	71.8	5	15.5	36.2	47.1

were petroleum ether > ether > chloroform > ethanol. They concluded that ethanol extracts of leaves and seeds seemed to produce mildest toxicity to rabbit eye skin and no toxicity to rabbit ear skin. However, in the present study, no such skin reaction was noted even after 24 h with 32% concentration of extract.

The optimized concentration of extract was applied on experimentally infested animals, and it was found that 86.4% of ticks of group II effected in 3 days while in group III (HT₁) and in group IV (HT₂), it was 65.9% and 71.8% within 5 and 6 days, respectively. A significantly high level of efficacy of *A. squamosa* was consistent with the observation of Chungsamarnyart et al. (1988) who tested the ethanolic extract of 44 plant species and a larvicidal action in the range of 90–100% was reported. However, the authors had not tested the acaricidal property of the plant extract against adult ticks, which is the most resistant stage of the parasite.

In in vivo study, the survived ticks after treatment were weighed individually and reared for egg production. There was a significant ($P<0.05$) reduction in the adult tick weight in all the treated groups as compared with control group. However, there was no significant difference in adult tick weight between treated groups (HT₁ and HT₂). The reduction in adult tick weight (DR%) has been recorded as 22.2%, 13.3%, 15.5% in groups II, III, and IV, respectively. In case of group II (CT), the reduction in adult tick weight may be due to the direct effect of the cypermethrin because it exerts primary action by targeting kinetics of sodium channels in nerve (Hart 1986). This action results in either repetitive discharge or membrane depolarization and subsequent death of target insect. Animals of groups III (HT₁) and IV (HT₂) recorded additional days of tick feeding as compared to animals of group II. However, there was significant ($P<0.05$) reduction in adult tick weight in groups III and IV in comparison to tick recovered from control group (group I). This finding suggests that there may be antifeedant components in both the plant extracts. Shankaram et al. (2006) of Council of Scientific and Industrial Research (CSIR) of India patented a novel compound isosquaomocine having insecticidal properties from *A. squamosa*. However, they did not study the acaricidal property of the compound.

There was no significant reduction in the RI of ticks from group III as compared to control, while a significant difference of 0.09 percentage in the RI of the ticks fed on group IV in comparison to control was recorded. The ticks from group II (CT) showed a reduction of 0.13% than the ticks of group I (control). This very high level of reduction in the RI may be due to direct effect of cypermethrin (Bicalho et al. 2001). Although cypermethrin has direct toxic effect on challenged ticks and not allowed ticks to feed on treated animals more than 3 days. The RI of ticks

fed on group IV animals, which fed on animals for two additional days than group II animals, did not differ significantly (0.04%) with RI of the ticks fed on group II, thus confirming the antifeedant property of neem seed extract.

The therapeutic efficacy (E%) of different treatments on the survived ticks was estimated as 76.1% in case of group II (CT) whereas 31.2% and 47.1% in case of groups III (HT₁) and IV (HT₂), respectively. In the present in vivo study, cypermethrin caused direct effect on 86.4% of the challenged ticks and showed 76.1% effectivity on future progeny of survived tick. This indicates that 23.9% ticks escaped the residual effect of cypermethrin. These ticks will be responsible for persistence of ticks in the environment and may contribute for resistance development to particular chemical acaricide.

Similarly, 28.2% and 34.1% of ticks escaped from direct effect of *A. indica* and *A. squamosa*, respectively, and will be responsible for environmental tick burden. However, there is a wide scope to reduce this burden by using same plant origin acaricide in subsequent application, as chances of development of resistance to herbal acaricide is comparatively less. The antifeedant, antiovipositional, and reduction in fecundity of ticks due to exposure of these plant extracts needs validation using more number of animals keeping the data generated in the present study as baseline. If validated successfully, these eco-friendly plant products may be utilized effectively in a strategic IPM format and may thus reduce the use of harmful chemical acaricide.

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